

SEED PRODUCTION, PROCESSING & SEED VILLAGE

LAST 5 PREVIOUS YEAR QUESTION (2019-2023)

- 1. Coconut was propagated by (2022)**
 - a. Seeds
 - b. Stems
 - c. Budding
 - d. None of the above

- 2. Which of the following is the only genetically modified crop approved for commercial cultivation by government of India? (2022)**
 - a. Bt brinjal
 - b. Flavr slavr tomato
 - c. Golden rice
 - d. Bt cotton

- 3. _____ is the initial seed obtained from the selected individual plants of a particular variety or parental line for the purpose of purifying and maintaining that variety or parental lines by the originating breeder and its further multiplication under his own supervision or the supervision of a qualified plant breeder. It is used to produce breeder seed and forms the basis of the total seed production. (2022 mains)**
 - a. Breeder seed
 - b. Certified seed
 - c. Nucleus seed
 - d. None of the above

- 4. What is the minimum purity percentage of certified seed? (2023)**
 - a. 70%
 - b. 99%
 - c. 96%
 - d. 99.9%

- 5. Foundation seeds are formed from which seeds in the stages of seed production by various state universities or government agencies? (2023)**
 - a. Nucleus seed

- b. Certified seed
- c. Breeders seed
- d. All of the above

6. Which one the following is a method for breaking dormancy of a seed?

- a. Soaking in a chemical solution
- b. Stratification
- c. Hot /cold water treatment
- d. All of the above

7. What is the colour tag used for the certified seed bag (2022)?

- a. Blue
- b. Yellow
- c. White
- d. None of the above

8. What is the colour tag for breeder's seed? (2021)

- a. Golden yellow
- b. Blue
- c. White
- d. None of above

Answers

1. A 2.D 3. C 4.B 5.C 6.D 7.A 8.A

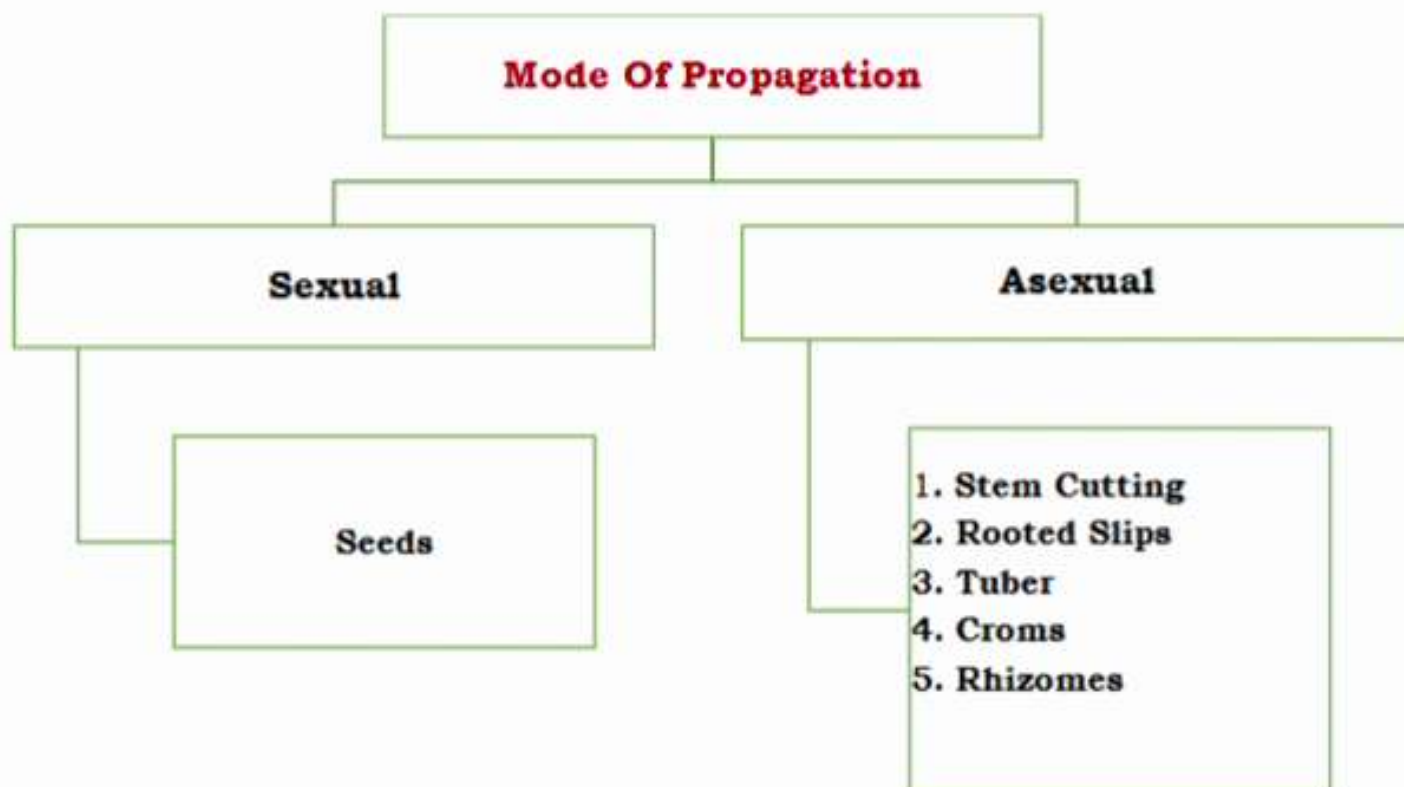
SEED SCIENCE

Seed is the basic and critical input in crop husbandry, which determines the expected dividends from all other inputs.

Seed Definition-Agriculturally seed is the unit of propagation and can be any part of the plant (zygotic seed or vegetative propagules) **which has the capability to regenerate into a new plant.**

Methods Of Propagation In Plants

Plants propagation or multiplication in plants is either by sexual or asexual means or by both. Plants reproduce **sexually by seeds** and **asexually by vegetative parts**. Most crop plants produce viable seeds which are used for sowing.



Flow Chart 1. Mode Of Propagation in Plants

Examples of propagation in various crops

1. **By seeds**- cereals, millets, pulses etc
2. **Stem cutting**- sugarcane & the stem cuttings of sugarcane is called as setts.
3. **Rooted slips** (rooted slips are basal two or three internodes of the stem with a few roots)-Napier grass, para grass, guinea grass.
4. **Tubers**-potato
5. **Chromes**- onion, garlic
6. **Grafting**, budding, & layering are used to propagated horticultural crops.

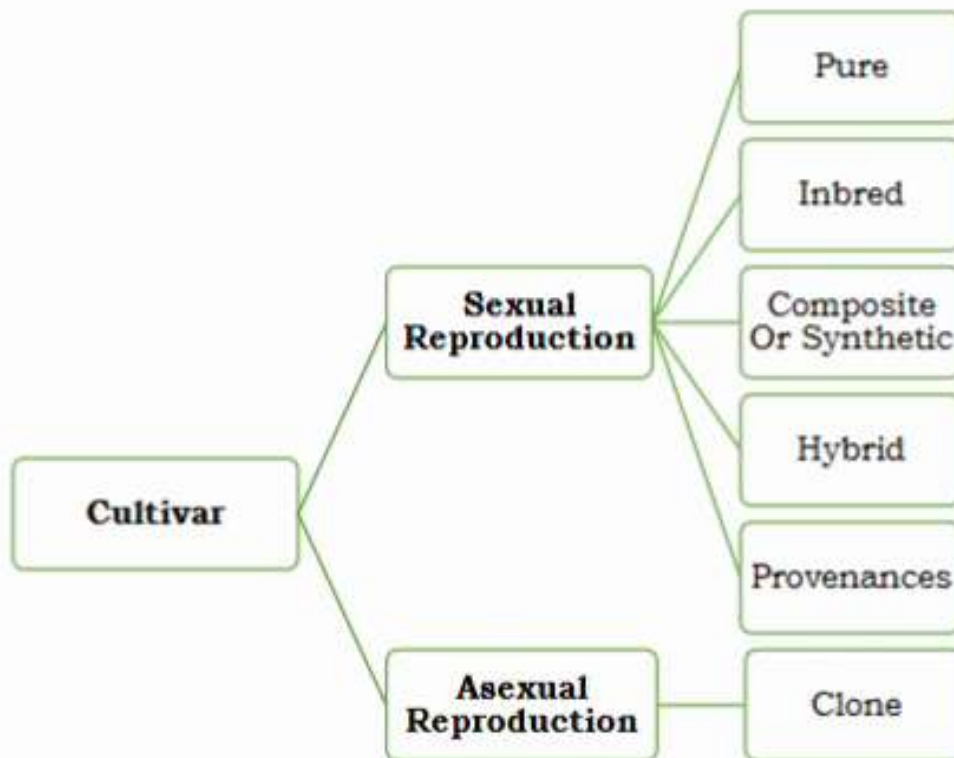
Some Terminologies Related to Seed Production

1. **Seed Germination**-Seed germination refers to the ability of a seed when planted under normal sowing conditions to give raise to a normal seedling.
2. **Seed Vigour**-The seed vigour refers to the sum total of all the attributes that gives effective plant stand in the field.
3. **Sexual Reproduction** – when there is a transfer of gametes (male or female) from one plant to the other plant of same species by the process of pollination is called as sexual reproduction
4. **Asexual Reproduction**- when the plant reproduces by its vegetative parts like leaves stems, tuber chrome etc is called as sexual reproduction. The offsprings of asexual reproduction are called as clone as they are genetically or physically similar to their parents.
5. **Cultivar**- word cultivar is used for the cultivated variety. a cultivar is a unique population of plants, but it is artificially maintained by human efforts and is named, many cultivars cannot continue to exist without human efforts **Ex sonara wheat, IR8 rice, samba mahsuri rice**
6. **Line**- line is population of seed propagated plants in which genetic variability is controlled and uniformity is maintained to a standard appropriate for that cultivar.

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7. **Pure line**- cultivars of self-pollinated plants which are propagated by seeds shows very little variation & complete resemblance to their parent plant.
8. **Inbred line**- these are true breeding cultivars resulted from enforced self-pollination of selected parent plants. Followed by continuous selection of desired type in succeeding generations.
9. **Synthetic cultivar** – it composed of plants produced by combining a number of genetically distinct but phenotypically similar lines which have been allowed to cross pollinate or self-pollinate naturally.
10. **Hybrid line**- a hybrid line is F1 generation of two inbreed lines.
11. **Provenances** – it is the geographic locality of the seed from which it was collected.
12. **Clone** – vegetative parts of asexually reproduced plants like cutting, division, graftings etc are called as clone.

Categories Of Cultivar



Flow Chart 2. Types Of Cultivars

Seed Quality –

Seed quality is a relative term and means the degree of excellence when compared to an acceptable standard.

Characteristics Of Seed Quality-

viability and vigour are the two important characters of the seed quality

1. **Viability-** Viability is expressed by the germination percentage which indicates the number of seedlings produced by a given number of seeds.
2. **Vigour-** The sum of all the properties of a seed which ensures its good health & germination rate it is difficult to measure but low germination percentage is associated with low vigour.

Low vigour seed may not be able to with stand unfavourable conditions, more the 85% seed germination is considered as good seed vigour.

Selection Of the Seed

For establishment of an optimum plant population of a crop in the field. The seed should satisfy the following conditions.

1. They should belong to the proper variety or strain of the crop which is proposed to be grown.
2. They should be free from mixtures of other seeds.
3. They should be mature, and well developed.
4. They should have high germinating capacity.
5. They should be free from pest and diseases.
6. They should germinate rapidly and uniformly.
7. Emergence should be prompt and early vegetative growth should be vigorous.

Seed Production

Seeds of an officially released and notified variety are sequentially produced through a limited generation system.

Agencies Involved In Seed Production

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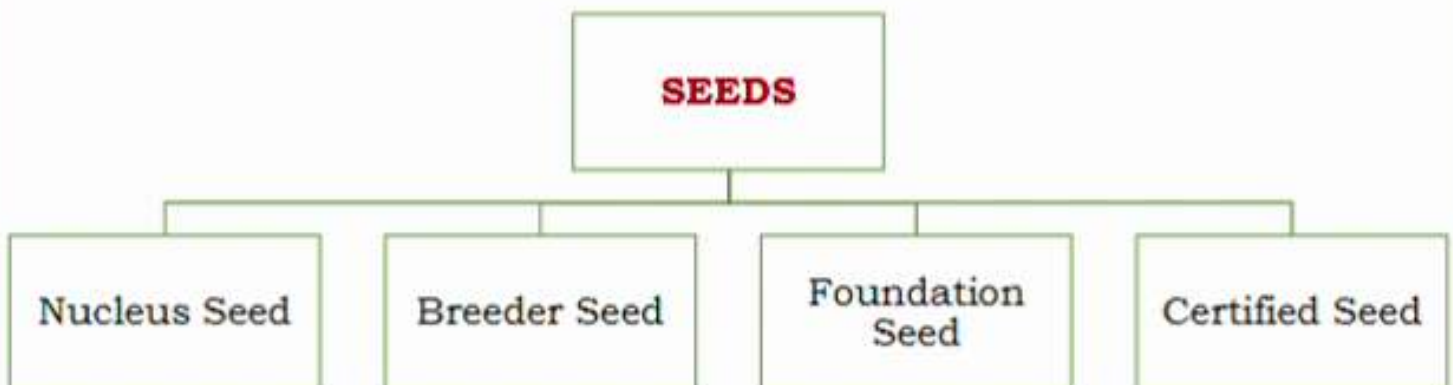
The seed production & supply system in India can be divided into formal and informal systems

1. **Formal seed production sector** – seed supplied through formal seed sector is characterised by **planned production, mechanized processing and inclusion of only identified or notified varieties** and a system of quality control.

Formal Seed Production & Supply System Include-

1. National govt agencies
 2. State govt agencies
 3. Govt assisted and other cooperatives
 4. MNC'S, private agencies, joint venture
 5. Centre & state universities
2. **Informal sector**-This system includes seed collected by the framers from their field, this system is traditional, & mainly operates at community level.

Types Of Seed In Seed Production System



Flow Chart 3- Types Of Seed

1. Nucleus seed-

- Nucleus seed is the initial **seed obtained from the selected individual plant of a particular variety or a parental line** for the purpose of purifying and maintaining that variety or parental line by the breeder for further multiplication.
- Nucleus seed are **used to produce breeder seed** and forms the basis of total seed production.
- Nucleus seeds have **100% purity percentage**



Truthful Labelled Seed

Nucleus seed are sometimes labelled as **Truthfully** labelled seed is also a **type of seed that is not certified but the labelling is done to indicate its quality or standard in a truthful manner.** The colour tag of this type of seed is **opal green** with size of colour tag 15 x 10 cm.

Image (ii) Truthful Seed Tag

2. Breeder Seed-

Breeder seeds are produced using nucleus seeds in the Research institutes or Universities under the supervision of a breeder. The entire production is supervised by the qualified plant breeder



Image (iii) – Tag of Breeder Seed

- It serves as a source for the **production of foundation seed.**

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- Breeder seed produced should meet all prescribed standards viz. genetic **purity (99.9% or more)**, physical purity (98% or more) Germination (as per crop) moisture content (less than 12 %).
- Breeder seed tags in **buff colour or Golden Yellow** and are signed by the concerned plant breeder and tagged to the breeder seed bags and size of tag is 12 × 6 cm.

3. Foundation Seed



Foundation seed

It is the **progeny of breeder seed** produced on the farms of State Agriculture Universities/Other government farms and State Seed Corporations.

Image (Iv)- Colour Tag Foundation Seed

- It is also produced by private seed producers.
- Its production can also be **taken up by the farmers by getting suitable breeder seeds.**
- It is **99.5 % genetically pure seed.**
- Colour tag of **foundation seed is white.**

4. Certified Seeds

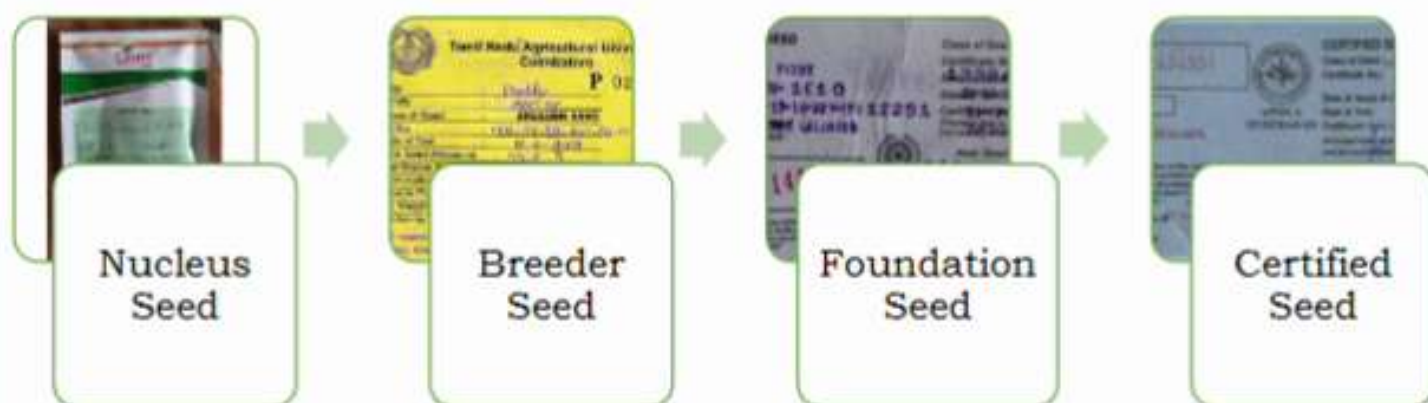


It is the **progeny of foundation seed** produced by registered seed growers under the supervision of seed certification agencies to maintain the seed quality as per minimum seed certification standards.

Image (V)- Colour Tag of Certified Seed

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- The production of certified seeds is taken up by the National and State Seed Corporation, private seed companies and also by farmers.
- It is **99% genetically pure seed**.
- **Colour tag of certified seed is blue, azar blue.**



Flow Chart – 4 Process of Production of Seed

Table 1. Characteristics Of Different Type of Seeds

<u>Characteristics</u>	<u>Nucleus Seed</u>	<u>Breeder Seed</u>	<u>Foundation Seed</u>	<u>Certified Seed</u>
<u>Produced From</u>	Taken from plants	Nucleus seed	Breeders seed	Foundation seed
<u>Purity Percentage</u>	100%	99.9%	99.5%	99%
<u>Colour Tag</u>	No colour	Golden yellow	White	Blue
<u>Size Of Colour Tag</u>	No size	12 x 6 cm	15 x 7.5 cm	15 x 7.5 cm

1. Registered seed



Registered seed shall be the progeny of foundation seed that is so handled as to maintain its genetic identity and purity according to standard specified for the particular crop being certified.

- A **purple colour certificate** is issued for this category of seed.
- These **seed do not form the parts of further seed production** as these seeds are only maintained for their genetic purity

Sowing Of Seed

Establishment of a **good stand is essential pre-requisite for attaining high yields**. Which depends on **seed bed preparation, time, depth & method** of sowing and seed treatment.

1. Seed Bed Preparation-

seed bed preparation **is the first operation for establishing of crop**. Generally, field is ploughed, harrowed, & then fine seed bed is prepared for sowing of crops

Ex Rice- field is puddled before transplanting of rice saplings.

2. Time Of Sowing-

Timely **sowing helps in achieving optimum utilization** of seasonal rainfall, reduces the incidence of pest & disease, and escape terminal drought.

Example

1. Cotton gives higher yield when sown in **June & July** as compared to august.
2. **Groundnut sown in July escapes** terminal drought & late leaf spot disease as compared to august sown crop.

Early sowing and delayed sowing of crops may not be advantageous because of the following reasons.

1. Gives lower yield

2. No efficient utilization of resources
3. More pest and weed incidence.
4. Reduced flowering.

3. Depth of Sowing -

Depth of sowing is **very important aspect** for establishing a good crop stand. uneven depth of sowing causes uneven crop stand which **causes a non-uniform crop stand** which creates a problem during harvesting of the crop.

- Thumb rule for **seed sow depth** is the depth should be 3 to 4 times of seed diameter.
- **Optimum depth** for most of the field crop is 3 to 5 cm.
- Small seeds like pearl millet, gingerly, or millets sown at 2 to 3cm depth
- Very small seed like tobacco sown at 1 cm depth.

4. Methods of Sowing a Seed

1. **Broadcasting**- seeds are sown by spreading the seeds uniformly over a well-prepared land by the hands and then cover the seed with the help of ploughing or planking. This is the most primitive way of sowing crops.
2. **Drilling**- seeds are sown in soil using the seed drills.
3. **Planting**- when seed or seed material of field crop are directly planted in the soil by manual labour or by machinery.
4. **Transplanting**- when the saplings of 20 to 30 days old field crop like rice are planted in the field from the nursery by the help of labour or machinery.

Examples of seed sowing

1. **Transplanting** – Rice, or vegetable crops.
2. **Groundnut & maize** are sown using country plough.
3. **Broad bed & furrow method** is adopted for establishing crops in **vertisols** (clayey soil).

5. Types of Sowing

1. Dry sowing-

Adopted in clayey soil where sowing operations are difficult to carry out after rainfall or irrigation so seeds are sown in dry soil before 7 to 10 days of anticipation of rainfall.

2. Wet sowing-

Most common method of sowing seed. Sowing is carried out after the rainfall or irrigation **when moisture is available up to 20mm depth in soil before sowing** of a crop.

Seed Germination

Seeds Germination:

It is resumption of **active growth of the embryo** after a period of dormancy in presence of favourable conditions viz. moisture, air, temperature, light, medium.

Germination Or Emergence of The Seed

Establishment of **good crops stand** ensures **successful growing of crops**. Poor crop stands results in **low plant population** and increases **severe weed problem** in the field which reduces yield.

There Can Be Several Reasons for Poor Crop Stand

1. Poor quality of seed with low germination
2. Improper seed bed preparation
3. Low moisture, low temperature at the time of sowing
4. Salinity and alkalinity of the soil.
5. High percentage of dormant seed or seed dormancy

Seed Dormancy

It is the condition **when seeds will not germinate** even when all the environmental conditions are in favour of germination.

Methods Of Breaking Seed Dormancy

1. Scarification:

Scarification is the process of **breaking, scratching, mechanically altering** or softening the seed **covering to make it permeable to water and gases**. Scarification should not exceed to the point at which seeds are injured. Scarified seeds are susceptible to the pathogenic organism, so scarification needs to be done at the time of sowing only.

2. Hot Water Treatment –

Drop the seeds into 4-5 times their volume of hot water with temperature ranging from 75 to 100°C and the seed soak in water for 12 to 24 hr the water will cool down gradually.

3. Acid Treatment:

Dry seeds are placed in containers and covered with **concentrated Sulphuric acid (H_2SO_4)** or **HCl twice the volume of the seed**. The duration of the treatment varies from 10 min to 6 hours depending upon the type seed or seed coat of that plant species.

4. Leaching-

The main purpose is to remove the germination inhibitor by soaking seeds under running water or by placing them in frequent water changes up to 12 to 24 hours.

5. Hormonal Treatment-

Treating seeds with gibberellic acid breaks the seed dormancy caused by physiological reasons. Seeds are **soaked in 500 ppm concentration for 12 hr**. ethylene & cytokinin is also used to break dormancy.

6. Priming-

Soaking seeds in **normal water for 24 to 48 hours** enhance germination by imbibition of water inside the seeds

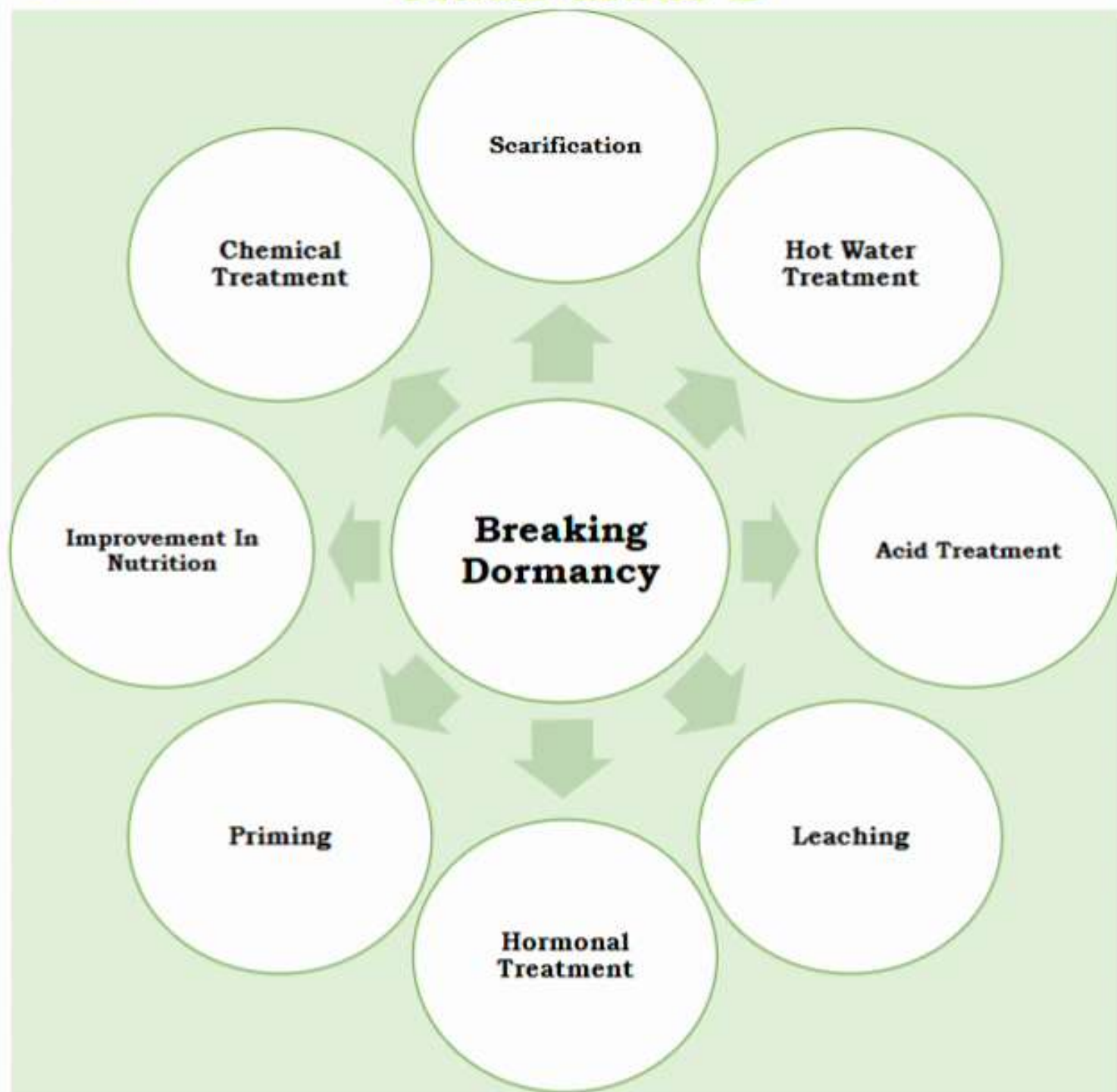
7. Improvement in nutrition –

Micronutrient deficiency of seeds can be overcome by enriching the seeds by spraying the micronutrient during crop season.

8. Chemical Treatment-

Thiourea is used to stimulate germination in some kinds of dormant seeds-

The **seeds are soaked in 0.5 – 3 per cent solution of thiourea for 3-5 minutes**. Afterwards seeds are rinsed with water and are sown in the field. Similarly, **potassium nitrate and sodium hypochlorite** also stimulate seed germination in many plant species.



Seed Storage

1. Storage at ambient temperature and humidity
2. Dry storage with control of moisture content but not temperature - Orthodox seeds will retain viability longer, when dried to low moisture content (4-8%) and then stored in a sealed container or in a room in which humidity is controlled
3. Dry storage with control of both moisture content and temperature - A combination of 4-8% moisture content and 0 to 5° temperature will maintain viability for 5 years or more.

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4. **Dry storage for long-term gene conservation** - Long-term conservation of gene resources of orthodox agricultural seeds is -18°C temperature and $5\pm 11\%$ moisture content
5. **Moist storage without control of moisture content** of temperature- Suitable for storage of recalcitrant seeds, for a few months over winter
6. **Cryopreservation**-It is also called as cryogenic storage. Seeds are placed in liquid nitrogen at -196°C . Seeds are actually placed into the gaseous phase of the liquid nitrogen -150°C for easy handling and safety

SEEDS VILLAGE

A village where **seeds production is done by the trained farmers** in cluster to produce high variety seeds.

Benefits

- Saving of time.
- **Reduction of cost.**
- Availability of **high-quality cost.**
- **Self-sufficiency and self-reliance** of the village.
- Increasing the **seed replacement rate.**

Seed Village Programme (Beej Gram Yojana)

Launched: 2014-15

Purpose: To upgrade the quality of farmer's saved seeds.

Objectives

- Increasing the **seed production.**
- Increasing the **seed replacement rate.**
- **Organizing seed production in cluster (or) compact area** replacing existing local varieties with new high yielding varieties.
- **Self-sufficiency and self-reliance of the village.**
- To train the **farmers about seed production technology**, use of quality seed along with improved **package of practices** for different seed crops.

Implementing agency:

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State Departments of Agriculture, State Agriculture Universities, **Krishi Vigyan Kendras**, **State Seeds Corporation**, **National Seeds Corporation**, State Farms Corporation of India (SFCI), State Seeds Certification Agencies, Department of Seed Certification.

One implementing agency will be identified for the area/locality and is to be authorized by the State Government.

Financial assistance

Under this programme financial assistance for distribution of foundation/certified seeds at 50% of seed cost for cereal crops and 60% for pulses, oilseeds, fodder and green manure crops is available for up to one acre per farmer.

The seed produced in these seed villages will have to be preserved/stored till the next sowing season.

Supply of Storage Bin

1. Assistance @ 33% subject to a maximum of Rs.3000/- for SC/ST farmers and @ 25% subject to maximum of Rs.2000 for other farmers for procuring seeds storage bin of 20 qtl. capacity.
2. Assistance @ 33% subject to maximum of Rs.1500/- to SC/ST farmers and @ 25% subject to maximum of Rs.1000/- for other farmers for making seeds storage bin of 10 qtl. capacity in the seed villages where seed village scheme is being implemented.

Monitoring of the Scheme

The scheme will be monitored by the Seeds Division of the Department of Agriculture and Cooperation. The implementing agencies will be required to send quarterly progress.

Phases

The present program of seed village scheme is having two phases

Seed production of different crops: The area which is suitable for raising a particular crop will be selected, and raised with a single variety of a kind.

Establishing seed processing unit: If the seeds are not processed and handled properly, all the past efforts in production may be lost. Thus, seed processing and packaging is a very important aspect of seed production.

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If the implementing agency feels that farmers are showing more interest they can form self-help-group (SHG) and take the assistance for setting up of seed processing unit under 25% back ended credit linked subsidy scheme.

Training: Assistance to train the farmers on seed production and seed technology @ Rs.15000/- for a group of 50-150 farmers.